

PIYUSH SHARMA

piyush.sharma10796@gmail.com | +1(510-509-0245) | www.linkedin.com/in/i-piyush

EDUCATION

Santa Clara University, Santa Clara, CA PhD in Electrical and Computer Engineering (HMI Lab, Robotics)	01/2026 - Present
San Francisco State University, San Francisco, CA Masters in Embedded Electrical & Computer Systems	05/2024
YMCA University of Science & Technology, India Bachelor of Technology, Electrical Engineering	05/2018

WORK EXPERIENCE

Embedded & Controls Engineer, Brillouin Energy, Emeryville, CA	09/2024 - Present
- Developed firmware in C/C++ and implemented HAL-based timer control, encoder input capture, and an in-house S-curve motion algorithm to significantly improve motor speed accuracy and positional stability. Validated motion control in collaboration with technicians and manufacturing team. - Led robot programming and optimization of ABB 6-axis robotic arm across multiple configurations; designed recipe-based control systems improving cycle times, operator usability, and scalability. Designed electrical schematics, wiring diagrams, and control panels for automation workcells. - Developed and validated a test bench using multiple protocols to collect, monitor, and log system data for cloud-based analytics. Collaborated with manufacturing, process engineering, mechanical engineering, and external vendors to integrate new tools and scale production processes. - Diagnosed and resolved critical hardware failures (plasma power supply transformer short, motor feedback issues, connector shorts), minimizing downtime through structured root-cause analysis. Deployed and automated two production lines end-to-end, encompassing PLCs, HMI, EtherNet/IP, and Keyence vision sensors; successfully established a full manufacturing process from the ground up.	
Firmware Engineer Intern, NeuroLeap Corp., San Jose, CA	01/2024 - 09/2024
- Led firmware development for low-power embedded systems using C/C++ with Zephyr RTOS. Designed MEMS sensor interface circuits (accelerometers, gyroscopes) using I2C and SPI protocols, enhancing system efficiency and reducing integration time by 20%. - Performed board bring-up, signal validation, and debugging using JTAG, oscilloscope, logic analyzer, and multimeter. Designed hardware schematics using EDA tools. Formulated verification plans for both component-level and system-level testing.	
Data Manager, HTC Logistics LLC., Union City, CA	01/2021 - 02/2023
Developed Python-based performance dashboards for a fleet of semi-trucks by aggregating sensor data across vehicles and surfacing real-time metrics on a centralized monitoring platform to support operational decision-making.	
Hardware Engineer Intern, Bhushan Power & Steel Ltd., India	06/2017 - 12/2017
- Developed firmware for board-level peripherals communicating over SPI, UART, and I2C using C++. Implemented state machine control logic on ESP32, STM32, and Arduino platforms. Programmed Allen-Bradley PLCs and servo controllers during routine plant operations. Implemented SCADA-based automation reducing operational downtime by 8%. - Designed electrical schematics and panel layouts in AutoCAD. Worked with embedded and industrial data transmission interfaces: I2C, SPI, UART/Serial, RS232, RS485, CAN Bus, and Modbus RTU.	

TECHNICAL SKILLS

Embedded & Firmware: C/C++, ARM Cortex-M4, STM32, FreeRTOS, Zephyr RTOS, HAL-based timer control, encoder input capture, S-curve motion algorithms, stepper motor control, PWM, CAN, UART, I2C, SPI, Modbus RTU

Robotics & Automation: ROS2, ABB 6-axis robots (RAPID), PLC programming, HMI design, EtherNet/IP, safety systems, vision sensors (Keyence), recipe-based control systems, SCADA

Tools & Debugging: GDB, Oscilloscope, Logic Analyzer, CAN Analyzer, JTAG, Wireshark, ABB RobotStudio, VS Code, GitHub, Ubuntu Linux, Schematics, wiring diagrams, control panel design, Altium, AutoCAD, Onshape, PCB design

PROJECTS

Autonomous Car using CAN Bus

Developed a self-navigating vehicle using ARM Cortex-M4 microcontrollers communicating over CAN Bus with FreeRTOS. Implemented non-preemptive real-time task scheduling with sensor fusion (accelerometers, gyroscopes, wheel encoders, GPS) under strict timing constraints, achieving 30% improvement in navigation accuracy.

Linux File System Design

Led a team of 4 to design a custom Volume Control Block file system with comprehensive metadata management and full file/directory operations.

ACHIEVEMENTS

Third place winner in Annual Research Competition, College of Science and Engineering at SFSU

2024